Binyi Gan

AI-Native Product Builder·Mobile 3D Creation ·PaintersGO

Not born different, but better at using the right tools.

PROJECT RESUME

PaintersGO — AI-Native Mobile 3D Creation Product Prototype

1. PROJECT OVERVIEW

PaintersGO is a mobile-first AI 3D creation prototype for Android. It lets users generate 3D models from text or images, paint and edit them directly on device, repair mesh geometry, collaborate in real time, and export production-ready GLB/STL assets — all within a single mobile workflow. Built over roughly 2–3 months as a fully independent, AI-assisted development effort, driven by the goal of making AI-generated 3D accessible to non-technical creators.

2. MY ROLE

Independently handled the full product lifecycle: concept, workflow design, requirement breakdown, AI-assisted development, multi-provider API integration, debugging, optimization, and acceptance testing. Used ChatGPT, Claude, and Gemini as development partners — while owning all architectural decisions, code review, bug reproduction, and quality control.

3. BUILT SCOPE

- Text-to-3D and image-to-3D generation via multiple AI providers
- Multi-provider orchestration — Meshy, Rodin, Tripo, Hunyuan (task submission, polling, download, local asset management)
- Mobile 3D editor: model loading, painting, erasing, saving, undo, and export
- GLB / STL asset handling with mobile performance optimization and polygon reduction pipeline
- FastAPI-based mesh repair service for watertight geometry
- WebSocket real-time multiplayer collaboration prototype (model, texture, and stroke sync)
- Batch export and Firebase Remote Config for remote feature management

4. TECHNICAL STACK

Mobile: Android, Kotlin, Jetpack Compose, WebView

3D / Frontend: Three.js, GLB / STL, model loading, material & texture handling

Backend: FastAPI, Node.js, WebSocket, REST API

AI Tools: ChatGPT, Claude, Gemini, Cursor, Android Studio AI workflow

Engineering: Git, GitHub, Gradle, Firebase Remote Config, Cloud Server / PM2

5. PRODUCT INSIGHT

The real bottleneck in AI 3D is not generation speed alone — it is asset readiness: topology quality, polygon budget management, format compatibility, mobile rendering performance, editing workflow fluidity, and printable output. Provider-generated high-poly assets led to a dedicated mobile optimization and post-processing pipeline, including polygon reduction, watertight repair, and localized paint control. The wider opportunity:
bridging AI 3D generation to physical manufacturing — enabling non-technical users to go from a text prompt to a printable,exportable asset.

6. AI-NATIVE DEVELOPMENT WORKFLOW

Requirement decomposition → multi-model solution generation → code integration → on-device testing → bug reproduction → iterative correction. This methodology enabled rapid cross-stack learning across Android, Three.js, FastAPI, and WebSocket, validating AI-assisted development as a reproducible, production-capable workflow.

7. LINKS

project website：https://paintersgo.top/ · GitHub：https://github.com/binyigan